

Unbiased AI Decision: Fairness Audit Report

Executive Summary

This report presents the results of a comprehensive fairness audit conducted on the provided dataset and/or machine learning model.

Dataset Bias Analysis

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{
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    "education",
    "education.num",
    "race",
    "sex",
    "income"
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      "43": 0.023647922361106844,
      "44": 0.022235189336936825,
      "45": 0.022542305211756396,
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  "Some-college": 0.22391818433094807,  
  "Bachelors": 0.16446055096587942,  
  "Masters": 0.05291606523141181,  
  "Assoc-voc": 0.042443413900064494,
```

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"11th": 0.03608611529129941,
"Assoc-acdm": 0.032769263843248055,
"10th": 0.02865391112066583,
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"Prof-school": 0.0176898743896072,
"9th": 0.01578575596572587,
"12th": 0.013298117379687356,
"Doctorate": 0.012683885630048217,
"5th-6th": 0.010226958631491662,
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"Preschool": 0.001566290961579804
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  "Black": 0.0959429992936335,
  "Asian-Pac-Islander": 0.03190933939375326,
  "Amer-Indian-Eskimo": 0.009551303706888609,
  "Other": 0.008322840207610331
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"sex": {
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  "Female": 0.33079450876815825
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"representation_imbalance": {
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"sex": {
  "max_min_ratio": 2.0230247887846997,
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```

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"percentage": 0.022542305211756396
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"percentage": 0.042443413900064494
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  "attribute": "education",
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  "group": "7th-8th",
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{
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{
  "attribute": "education",
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"group": "1st-4th",
"percentage": 0.005159546696968767
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{
  }
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{
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"percentage": 0.03190933939375326
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{
"attribute": "race",
"group": "Amer-Indian-Eskimo",
"percentage": 0.009551303706888609
},
{
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}
],
"fairness_metrics": {
"error": "No target column found"
}
}
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Recommendations

Consider collecting more data for underrepresented groups.

Address imbalance in age attribute through data collection or augmentation.

Address imbalance in education attribute through data collection or augmentation.

Address imbalance in education.num attribute through data collection or augmentation.

Address imbalance in race attribute through data collection or augmentation.